

Data Sheet for Exp-6

Observation and Manipulation:

Pitch of the screw gauge, $P = 1 \text{ (mm)}$

Total no. of circular division of the screw gauge, $N = 100$

Least count of the screw gauge $= \frac{P}{N} = 0.01 \text{ (mm)} = 0.001 \text{ (cm)}$

Table 1: For the determination of resistance of the experimental wire:

No. of Obs.	Known resistance = P (ohm)	Position of the resistance, P	Distance of the null point = l (m)	Unknown resistance (ohm)	Resistance, Q (ohm)	Average resistance, Q (ohm)
1.	2	Left gap	80	0.5	0.51	0.53
2.	3		86	0.49		
3.	4		88	0.54		
1.	2	Right gap	21	0.53	0.55	
2.	3		16	0.57		
3.	4		12	0.54		

Table 2: For the determination of radius of the experimental wire:

No. of Obs.	Main scale reading = M (mm)	Circular scale reading = V	Least count = K (mm)	Fractional part, $F = V \times K$ (mm)	Total reading, $d' = M + F$ (mm)	Average reading, d (mm)	Radius, $r = d/2$ (mm)	Average radius, r (m)
1.	0	36	0.01					
2.	0	41	0.01					
3.	0	37	0.01					

Table 3: For the determination of specific resistance of experimental wire:

Length of the wire, L (m)	Total resistance of the wire, Q (ohm)	Cross-sectional area of the wire, $A = \pi r^2$ (m^2)	Specific resistance, $\rho = \frac{Q\pi r^2}{L}$ (Ωm)
0.23	0.53		

Calculation:

$$\rho = \frac{Q\pi r^2}{L} = \dots \dots \dots (\Omega m)$$

(The answer will be around - $2.34 \times 10^{-6} \Omega m$)

Result:

Specific resistance of the experimental wire, $\rho = \dots \dots \dots (\Omega m$